

Chapter 06

Mining frequent Patterns, Associations, and Correlations: Basic Concepts and Methods.

→ Frequent pattern mining searches for recurring relationships in a given dataset.

If we think of the universe as the set of items available at the store, then each item has a Boolean variable representing the presence or absence of that item. These patterns can be represented in the form of association rules.

For example, the information who purchase computers also tend to buy antivirus software at the same time is represented in the following association rules:

computer $\Rightarrow$ antivirus software [support=2%, confidence=60%].

Here support reflect usefulness and confidence reflect certainty. A support of 2% means that 2% of all the transactions under analysis show that computer and antivirus ~~for~~ soft purchased together. A confidence 60% means that 60% of the customers who purchased a computer also bought the software.

Let $I = \{I_1, I_2, \dots, I_m\}$ be an itemset. An association rule is an implication of the form $A \Rightarrow B$, where $A \subset I$, $B \subset I$, $A \neq \emptyset$, $B \neq \emptyset$, and $A \cap B = \emptyset$.

The rule $A \Rightarrow B$ has support s in the transaction set D , where s is the percentage of transactions in D that contain $A \cup B$. And the rule $A \Rightarrow B$ has confidence c in the transaction set D , where c is the percentage of transactions in D that contain $P(B|A)$. That is,
$$\text{support}(A \Rightarrow B) = P(A \cup B); \text{confidence}(A \Rightarrow B) = P(B|A).$$

$$\text{confidence}(A \Rightarrow B) = P(B|A) = \frac{\text{support}(A \cup B)}{\text{support}(A)} = \frac{\text{support_count}(A \cup B)}{\text{support_count}(A)}$$

In general, association rules mining can be viewed as a two-step process:

- ① Find all frequent itemsets;
- ② Generate strong association rules from the frequent itemsets:

* To improve the efficiency of the level-wise generation of frequent itemsets, an important property called the Apriori property is used to reduce the search space.

* Apriori Algorithm: finding frequent itemsets by Confined Candidate Generation.

Apriori is a seminal algorithm for mining frequent itemsets for Boolean association rules. This algorithm uses prior knowledge of frequent itemset properties. It employs an interactive approach known as a level-wise search.

To improve the level efficiency of the level-wise generation of frequent itemsets, an important property called the Apriori property is used to reduce the search space.

* **Apriori property**: All nonempty subsets of a frequent itemset must also be frequent.

↳ **Antimonotonicity** - the sense that if a set cannot pass a test, all of its supersets will fail in the same test as well.

* How is the apriori property used in the algorithm?
⇒ A two-step process is followed, consisting of join & prune actions.

□ **Join step**: To create a pool of potential frequent k -itemsets (C_k), the algorithm joins the frequent itemsets from the previous level (L_{k-1}) with themselves. It takes two different $(k-1)$ -itemsets and combines them into a new k -itemset only if they share the exact same first $k-2$ items.

□ **Prune step**: This is where the Apriori property is applied to shrink the candidate pool. The algorithm checks every $(k-1)$ subset of the newly joined candidates. If a candidate contains any subset that is not present in L_{k-1} , it is instantly deleted.

P	$\{P1, S1\}$
S	$\{P1, S1\}$
S	$\{P1, S1\}$

S	$\{P1, S1\}$
S	$\{P1, S1\}$
S	$\{P1, S1\}$

S	$\{P1, S1\}$
S	$\{P1, S1\}$
S	$\{P1, S1\}$

Example 6.3 : Apriori

- Initial Setup:
- ⇒ The dataset contain 9 transactions, meaning $|D| = 9$.
 - ⇒ The minimum support count is 2, which corresponding to a relative support of 22%.

Iteration 1: finding 1-itemsets:

↳ Candidate Generation (C_1): The algorithm first scans all 9 transactions to count the occurrences of each individual item.

↳ filtering (L_1): Items meeting the minimum support count of 2 become the set of frequent 1-itemsets.

C_1

Itemset	Sup. Count
{1}	6
{2}	7
{3}	6
{4}	2
{5}	2

L_1

Itemset	Sup. Count
{1}	6
{2}	7
{3}	6
{4}	2
{5}	2

Scan D for count of each candidate item → Compare candidate support count with minimum support count.

Iteration 2: finding 2-itemsets:

C_2

Itemset	Sup. Count
{1, 2}	4
{1, 3}	4
{1, 4}	1
{1, 5}	2
{2, 3}	4
{2, 4}	2
{2, 5}	2
{3, 4}	0
{3, 5}	1
{4, 5}	0

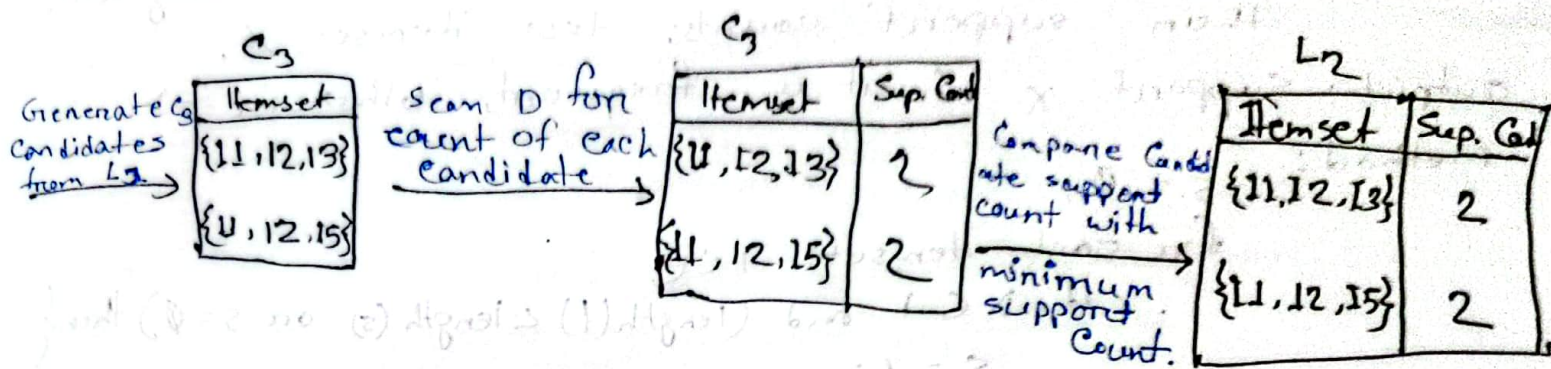
L_2

Itemset	Sup. Count
{1, 2}	4
{1, 3}	4
{1, 5}	2
{2, 3}	4
{2, 4}	2
{2, 5}	2

Generate C_2 candidate from L_1 → Scan D for count of each candidate → Compare candidate support count with minimum support count.

Generate C_2 candidate from

Iteration 3: finding 3-itemsets:



Ex 69 Generating association rules:

The data contain frequent itemset $X = \{11, 12, 15\}$.

What are the association rules that can be generated from X ?

Table 6.1 Transaction Data for an AllElectronics Branch

TID	List of item-IDs
T100	11 12, 15
T200	12, 14
T300	12, 13
T400	11, 12, 14, 15
T500	11, 13
T600	12, 13
T700	11, 13
T800	11 12 13 15
T900	11, 12, 13

The nonempty subsets of X are $\{11\}$, $\{12\}$, $\{15\}$, $\{11, 12\}$, $\{11, 15\}$, $\{12, 15\}$

The resulting association rules are as shown below, each listed with its confidence

$\{11, 12\} \Rightarrow 15$, confidence = $2/4 = 50\%$

$\{11, 15\} \Rightarrow 12$, confidence = $2/2 = 100\%$

$\{12, 15\} \Rightarrow 11$, confidence = $2/2 = 100\%$

$11 \Rightarrow \{12, 15\}$, confidence = $2/6 = 33\%$

$12 \Rightarrow \{11, 15\}$, confidence = $2/7 = 28\%$

$15 \Rightarrow \{11, 12\}$, confidence = $2/2 = 100\%$

If the minimum confidence threshold is, say 70% then only the 2nd, 3rd & last rules are output.